

Vacuum Charge Furnace PR-170029 — Final Cost Summary

Cenovus Lima, OH · Job USA26046 · WO 82036776 · Quotation DSP#26013 · 19 September 18:00 – 30 September 01:30, 2026 · 21 shifts · Service receipts 7077–7100

QUOTED \$107,200 134 hours	FINAL \$189,575 247.5 hours	VARIANCE +\$82,375 +77%	PASSES COMPLETED 8 of 8 decoked and inspected
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Scope completed

- All eight passes decoked through the convection and all five radiant segments, 3,595 ft of convection and 3,459 ft of radiant tube.
- Clean IDs confirmed by field measurement at 4.0", 3.8", 5.7", 7.8", 9.75" and 11.75", matching the coil drawings.
- All eight passes smart pigged. Pass 1–3 data approved 25 September; passes 4 and 6 on 26 September; passes 5, 7 and 8 completed 28 September.
- Filtration run throughout on filter press 1.
- Heater dewatered, launchers and hoses removed, filter presses cleaned at the wash pad, and equipment demobilised from the unit.

Basis. DSP#26013 prices this heater at 134 hours × **\$800 per hour** — pumping unit \$500, support trailer \$30, filter press \$245 and crew truck \$25, all four on every quoted hour, at the rates on the Cenovus-approved rate sheet. Labour, per diem, DEF and materials are quoted at project level across all four heaters and are invoiced separately.

Pumping unit, by task

TASK	QUOTED HRS	ACTUAL HRS	Δ HRS	QUOTED	ACTUAL	Δ	DETAIL
Rig-in	6	14	+8	\$3,000	\$7,000	+\$4,000	Rig-in could not be completed in a single pass. It was worked in four separate periods of 5, 2, 4 and 3 hours as access to the launcher connections allowed, each restart carrying its own set-up.
Pigging	66	63	-3	\$33,000	\$31,500	-\$1,500	All eight passes decoked within the quoted allowance. The coil cleaned to size on the planned progression, with minimal deposit found in the larger radiant bores.
Smart pig support	38	74	+36	\$19,000	\$37,000	+\$18,000	Pumping unit time supporting the inspection contractor. Inspection was accepted section by section rather than by complete pass, so passes were re-entered to collect the outstanding diameters. The 4", 6" and 8" sections each required repeat runs; the field record of 24 September notes data loss on pass 3 and on the 4" of passes 1 and 2.
Rig-over	18	8	-10	\$9,000	\$4,000	-\$5,000	Two rig-overs were required rather than the phased moves the quotation assumed, saving ten hours against plan.
Rig-out	6	36.5	+30.5	\$3,000	\$18,250	+\$15,250	Dewatering of all eight passes, removal of the 6" and 12" launchers and hoses, the filter press dropped and cleaned at the wash pad, pigs and crates loaded, and equipment demobilised from the unit and the facility. The final shift ran 18:00 to 01:30 on 29–30 September; all units are billed to that finish, not a full shift.
Productive total	134	195.5	+61.5	\$67,000	\$97,750	+\$30,750	

Equipment

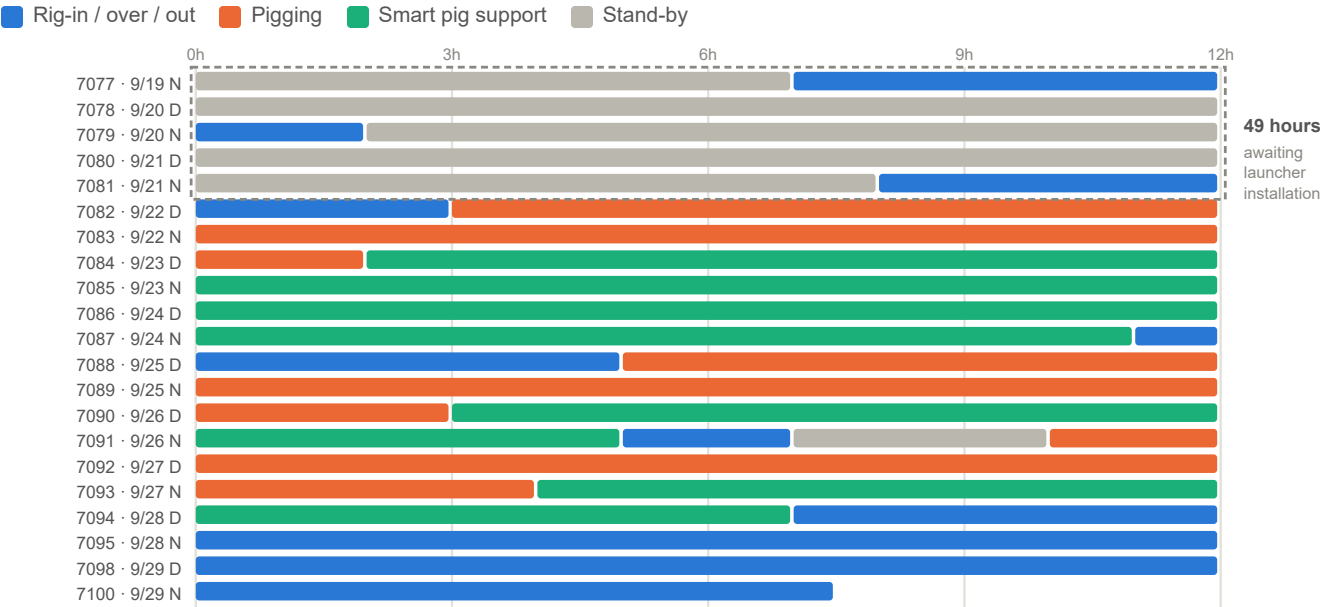
LINE	RATE	QUOTED HRS	ACTUAL HRS	QUOTED	ACTUAL	Δ	DETAIL
Support trailer 6	\$30	134	240.5	\$4,020	\$7,215	+\$3,195	On site from 01:00 on 20 September through demobilisation.
Filter press 1	\$245	134	235.5	\$32,830	\$57,698	+\$24,868	Filtration ran for the full duration of the job. The largest single equipment line, and the one most affected by the extended schedule.
Crew trucks	\$25	134	244.5	\$3,350	\$6,113	+\$2,763	One crew truck per shift across a 21-shift job.
Equipment total		134	—	\$40,200	\$71,025	+\$30,825	

Stand-by

Fifty-two hours were booked to stand-by at the rate sheet's \$400, a category the quotation carries no allowance for. Every one of those hours is recorded on a dated field ticket as waiting on the **12" radiant-outlet launchers to be installed by facility pipefitters**. Forty-nine were spent before the launchers were first hung on the night of 21 September, and a further three on 26 September waiting for them to be moved to passes 5, 7 and 8. The requirement is named on every ticket across that run.

PERIOD	HOURS	COST	DETAIL
19–21 September	49	\$19,600	Pumping unit and crew on site and available. Convection launchers and hoses were connected during this period as access allowed; the coil could not be entered until the radiant-outlet launchers were in place.
26 September	3	\$1,200	Waiting for the 12" launchers to be relocated to passes 5, 7 and 8 before the final three passes could be filled.

Stand-by total **52 \$20,800**



Summary



Quoted — 134 hours at \$800	\$107,200
Pumping unit, productive work above quote	+\$30,750
Equipment across the extended schedule	+\$30,825
Stand-by, 52 hours	+\$20,800
Final — 247.5 hours	\$189,575

Prepared from daily service receipts 7077–7100. Every hour shown is as recorded on a dated field ticket; no figure is estimated or allocated across tasks. Costs are hours at the rates on the Cenovus-approved rate sheet reproduced in DSP#26013. Labour, per diem, DEF and materials are quoted at project level and invoiced separately. USA DeBusk, LLC.